



Datu struktūras(English)(1),23/24-P Final online exam

Started on Monday, 3 June 2024, 9:19 PM**State** Finished**Completed on** Monday, 3 June 2024, 9:48 PM**Time taken** 29 mins 29 secs**Marks** 26.00/30.00**Grade** 8.67 out of 10.00 (86.67%)

Question 1

Complete

Mark 1.00 out of 1.00

LinkedList in Data Structures refers to the process of transforming a given key to another value. It involves mapping data to a specific index in a hash table using a hash function that enables fast retrieval of information based on its key. The transformation of a key to the corresponding value is done using a Hash Function and the value obtained from the hash function is called LinkedList Code.

☐ True☒ False

Question 2

Complete

Mark 1.00 out of 1.00

The Graph In The Data Structure

It is a type of linear data structure that consists of a root or child connected by edges or arcs to enable storage or retrieval of data. roota may be directed or undirected.

☐ True☒ False

Question 3

Complete

Mark 1.00 out of 1.00

Common operations performed on arrays

- Traversal: Visiting each element of an array in a specific order (e.g., sequential, reverse).
- Insertion : Adding a new element to an array at a specific index.
- Deletion : Removing an element from an array at a specific index.
- Searching : Finding the index of an element in an array.

☒ True

☐ False

Question 4

Complete

Mark 1.00 out of 1.00

Need of asymptotic analysis of an algorithm

Asymptotic analysis is the technique of determining an algorithm's running time in mathematical units to determine the program's limits, also known as "run-time performance." The purpose is to identify the best case, worst case, and average-case times for completing a particular activity.

While not a deep learning training technique, Asymptotic analysis is an essential diagnostic tool for programmers to analyze an algorithm's efficiency rather than its correctness.

☒ True

☐ False

Question 5

Complete

Mark 1.00 out of 1.00

Stack refers to the process of generating a fixed-size output from an input of variable size using the mathematical formulas known as stack functions. This technique determines an index or location for the storage of an item in a data structure.

☐ True

☒ False



Question 6

Complete

Mark 1.00 out of 1.00

Guess the output for the function to search a given element x in arr[]

```
Input : arr[] = {10, 20, 80, 30, 60, 50,  
                110, 100, 130, 170}  
        x = 110;
```

- ☐ a. Element x is present at index 0
- ☐ b. Element x is present at index 7
- ☒ c. Element x is present at index 6
- ☐ d. Element x is present at index 1
- ☐ e. Element x is present at index 4

Your answer is correct.



Question 7

Complete

Mark 1.00 out of 1.00

Guess the output

```
it('should be possible to convert stack to array', () => {  
  const stack = new Stack();  
  
  expect(stack.peek()).toBeNull();  
  
  stack.push(1);  
  stack.push(2);  
  stack.push(3);  
  
  expect(stack.toArray()).toEqual([3, 2, 1]);  
});
```

- ☐ a. 3 0 1
- ☐ b. 1 2 3
- ☐ c. 3 3 3
- ☒ d. 3 2 1

Your answer is correct.

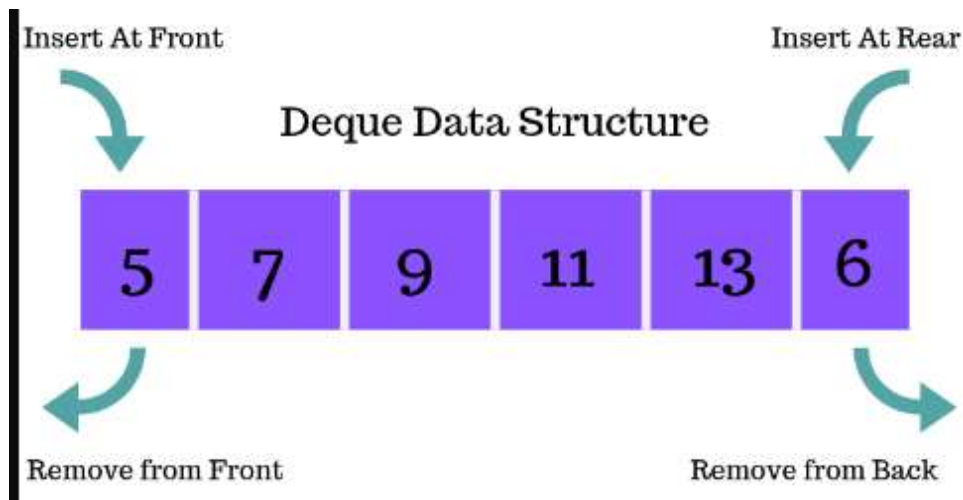


Question 8

Complete

Mark 1.00 out of 1.00

A **deque** can be thought of as an array of items, but with one important difference: Instead of pushing and popping items off the end to make room, deques are designed to allow items to be inserted at either end. This property makes deques well-suited for performing tasks such as keeping track of inventory, scheduling tasks, or handling large amounts of data.



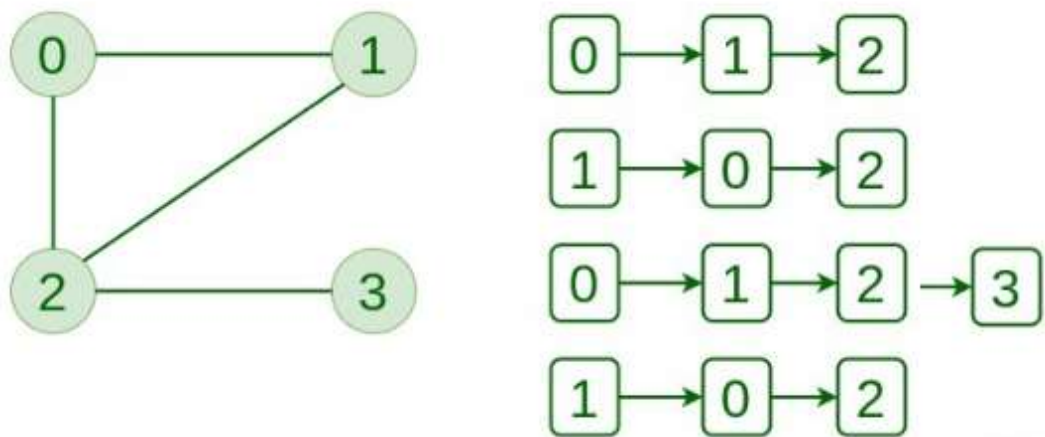
- ☒ True
- ☐ False

Question 9

Complete

Mark 0.00 out of 1.00

Adjacency List In Graph



- ☒ True
- ☐ False



Question 10

Complete

Mark 1.00 out of 1.00

The different types of arrays:

One-dimensional array: A one-dimensional array stores its elements in contiguous memory locations, accessing them using a single index value. It is a linear data structure holding all the elements in a sequence.

Two-dimensional array: A two-dimensional array is a tabular array that includes rows and columns and stores data. An $M \times N$ two-dimensional array is created by grouping M rows and N columns into N columns and rows.

Three-dimensional array: A three-dimensional array is a grid that has rows, columns, and depth as a third dimension. It comprises a cube with rows, columns, and depth as a third dimension. The three-dimensional array has three subscripts for a position in a particular row, column, and depth. Depth (dimension or layer) is the first index, row index is the second index, and column index is the third index.

- ☒ True
- ☐ False

Question 11

Complete

Mark 1.00 out of 1.00

A tree data structure is a hierarchical structure that is used to represent and organize data in a way that is easy to navigate and search. It is a collection of nodes that are connected by edges and has a hierarchical relationship between the nodes.

- ☒ True
- ☐ False

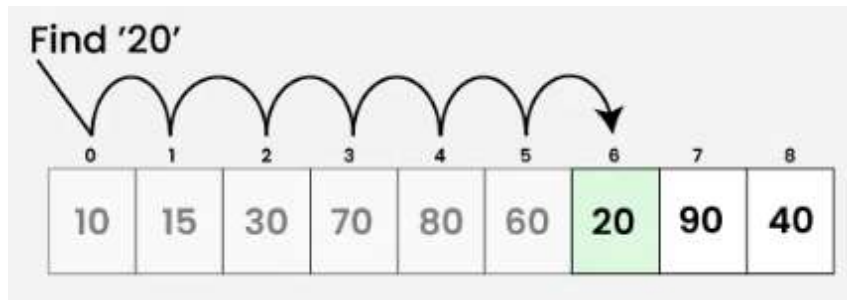


Question 12

Complete

Mark 1.00 out of 1.00

Linear Search is defined as a sequential search algorithm that starts at one end and goes through each element of a list until the desired element is found, otherwise, the search continues till the end of the data set.



- ☒ True
- ☐ False

Question 13

Complete

Mark 1.00 out of 1.00

A stack is an abstract data type that specifies a linear data structure or an ordered list, using the First In First Out (FIFO) operation to access elements. Insert operations can be performed only at one end called REAR and delete operations can be performed only at the other end called FRONT.

- ☐ True
- ☒ False

Question 14

Complete

Mark 1.00 out of 1.00

Circular Linked List: A circular linked list is a data structure that allows for two-way data access such that each node in the list points to the next node in the list and also points back to its previous node.

- ☐ True
- ☒ False



Question 15

Complete

Mark 1.00 out of 1.00

Guess the hash table logic

```
it('should track actual keys', () => {  
  const hashTable = new HashTable(3);  
  
  hashTable.set('a', 'sky-old');  
  hashTable.set('a', 'sky');  
  hashTable.set('b', 'sea');  
  hashTable.set('c', 'earth');  
  hashTable.set('d', 'ocean');  
  
  expect(hashTable.getKeys()).toEqual(['a', 'b', 'c', 'd']);  
  expect(hashTable.has('a')).toBe(true);  
  expect(hashTable.has('x')).toBe(false);  
  
  hashTable.delete('a');  
  
  expect(hashTable.has('a')).toBe(false);  
  expect(hashTable.has('b')).toBe(true);  
  expect(hashTable.has('x')).toBe(false);  
});
```

- ☐ a. After hash tabel delete!! 'a' is false and 'a' is false
- ☐ b. After hash tabel delete!! 'b' is false and 'a' is false
- ☒ c. After hash tabel delete!! 'a' is false and 'x' is false
- ☐ d. After hash tabel delete!! 'c' is false and 'b' is false
- ☐ e. After hash tabel delete!! 'a' is false and 'a' is false

Your answer is correct.



Question 16

Complete

Mark 1.00 out of 1.00

A **stack is a complete** stack tree data structure that satisfies the stack property: for every node, the value of its children is less than or equal to its own value. stack are usually used to implement priority queues, where the smallest (or largest) element is always at the root of the tree.

- ☐ True
- ☒ False

Question 17

Complete

Mark 1.00 out of 1.00

A Binary Tree Data Structure is a hierarchical data structure in which each node has at most three children, referred to as the left child, centre child and the right child.

- ☐ True
- ☒ False

Question 18

Complete

Mark 1.00 out of 1.00

Different operations available in the queue

- **enqueue:** This adds an element to the rear end of the queue. Overflow conditions occur if the queue is full.
- **dequeue:** This removes an element from the front end of the queue. Underflow conditions occur if the queue is empty.
- **isEmpty:** This returns true if the queue is empty or else false.
- **rear:** This returns the rear-end element without removing it.
- **front:** This returns the front-end element without removing it.
- **size:** This returns the size of the queue

- ☒ True
- ☐ False



Question 19

Complete

Mark 0.00 out of 1.00

Components of a Graph in DS:

Vertices: Vertices are the fundamental units of the graph. Sometimes, vertices are also known as vertices or nodes. Every node/vertex can be labelled or unlabelled.

Edges: Edges are drawn or used to connect two nodes of the graph. It can be an ordered pair of nodes in a directed graph. Edges can connect any two nodes in any possible way. There are no rules. Sometimes, edges are also known as arcs. Every edge can be labelled/unlabelled.

- ☐ True
- ☒ False

Question 20

Complete

Mark 1.00 out of 1.00

Types of Data Structures?

- **Linear Data Structure:** A data structure that includes data elements arranged sequentially or linearly, where each element is connected to its previous and next nearest elements, is referred to as a linear data structure. Arrays and linked lists are two examples of linear data structures.
- **Non-Linear Data Structure:** Non-linear data structures are data structures in which data elements are not arranged linearly or sequentially. We cannot walk through all elements in one pass in a non-linear data structure, as in a linear data structure. Trees and graphs are two examples of non-linear data structures

- ☒ True
- ☐ False



Question 21

Complete

Mark 0.00 out of 1.00

Asymptotic Notations are programming languages that allow you to analyze an algorithm's running time by identifying its behaviour as its input size grows. This is also referred to as an algorithm's growth rate.

Asymptotic notations are classified into three types:

- Big-Oh (O) notation
- Big Omega (Ω) notation
- Big Theta (Θ) notation

☐ True

☒ False

Question 22

Complete

Mark 1.00 out of 1.00

Dynamic memory management is a technique in which storage units are allocated based on the requirements continuously. Using dynamic memory allocations, individual data structures can be either stored separately or combined to form entities called composites. These composites can be worked on when required.

☒ True

☐ False

Question 23

Complete

Mark 1.00 out of 1.00

A Queue Data Structure is a fundamental concept used for storing and managing data in a specific order. It follows the principle of "Last in, First out" (LIFO).

☐ True

☒ False



Question 24

Complete

Mark 1.00 out of 1.00

Guess the logic for linked-list

```
it('should reverse linked list', () => {
  const linkedList = new LinkedList();

  // Add test values to linked list.
  linkedList
    .append(1)
    .append(2)
    .append(3);

  expect(linkedList.toString()).toBe('1,2,3');
  expect(linkedList.head.value).toBe(1);
  expect(linkedList.tail.value).toBe(3);

  // Reverse linked list.
  linkedList.reverse();
  expect(linkedList.toString()).toBe('3,2,1');
  expect(linkedList.head.value).toBe(3);
  expect(linkedList.tail.value).toBe(1);
});
```

- ☐ a. Reverse LinkedList Head is 1 and the trail is 2
- ☒ b. Reverse LinkedList Head is 3 and the trail is 1
- ☐ c. Reverse LinkedList Head is 1 and the trail is 3
- ☐ d. Reverse LinkedList Head is 1 and the trail is 1
- ☐ e. Reverse LinkedList Head is 3 and the trail is 1

Your answer is correct.



Question 25

Complete

Mark 1.00 out of 1.00

Tree Data Structure is a linear data structure in which a collection of arrays known as array nodes are connected via array edges such that there exists exactly one path between any two array nodes.

- ☐ True
- ☒ False

Question 26

Complete

Mark 1.00 out of 1.00

Guess the output for queue in FIFO

```
it('should dequeue from queue in FIFO order', () => {  
  const queue = new Queue();  
  
  queue.enqueue('mango');  
  queue.enqueue('kiwi');  
  queue.enqueue('kiwi');  
  
  expect(queue.dequeue()).toBe('mango');  
  expect(queue.dequeue()).toBe('kiwi');  
  expect(queue.dequeue()).toBe('kiwi');  
  expect(queue.dequeue()).toBeNull();  
  expect(queue.isEmpty()).toBe(true);  
});
```

- ☒ a. mango kiwi kiwi
- ☐ b. kiwi kiwi kiwi
- ☐ c. mango mango mango
- ☐ d. mango kiwi mango
- ☐ e. none of above

Your answer is correct.

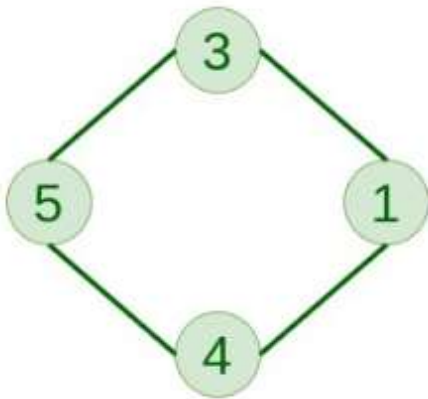


Question 27

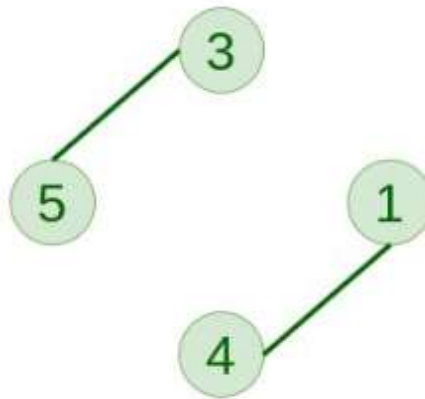
Complete

Mark 1.00 out of 1.00

Complete Graph and Regular Graph



2-Regular



Complete Graph

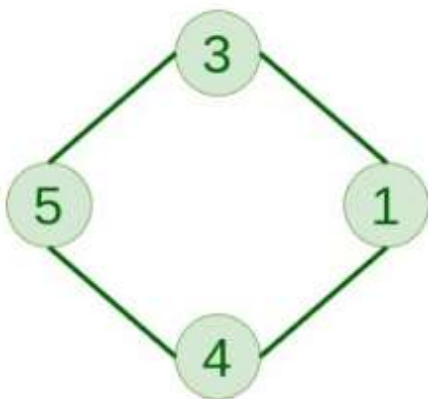
- ☐ True
- ☒ False

Question 28

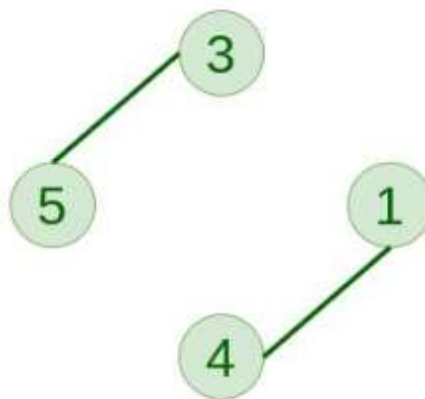
Complete

Mark 1.00 out of 1.00

Complete Graph and Regular Graph



2-Regular



Complete Graph

- ☐ True
- ☒ False



Basic Operations on Queue Data Structure

1. Enqueue (Insert): Adds an element to the rear of the queue.

2. Dequeue (Delete): Removes and returns the element from the front of the queue.

3. Peek: Returns the element at the front of the queue without removing it.

4. Empty: Checks if the queue is empty.

5. Full: Checks if the queue is full.

True

False

Differentiate between stack and queue

Stack	Queue
Stack is a linear data structure where data is added and removed from the top.	Queue is a linear data structure where data is ended at the rear end and removed from the front.
Stack is based on LIFO(Last In First Out) principle	Queue is based on FIFO(First In First Out) principle
Insertion operation in Stack is known as push.	Insertion operation in Queue is known as eneque.
Delete operation in Stack is known as pop.	Delete operation in Queue is known as dequeue.
Only one pointer is available for both addition and deletion: top()	Two pointers are available for addition and deletion: front() and rear()
Used in solving recursion problems	Used in solving sequential processing problems

- True
- False

